

Executive Report: Adult Census Income Prediction

Capstone

Business Goal

The objective of this project was to develop a complete machine learning solution to identify individuals whose annual income exceeds **\$50,000** using the UCI Adult Census Income dataset. The primary business objective was to **maximize Precision**, ensuring that individuals predicted as high-income are highly likely to belong to the positive class. This objective is particularly relevant for applications such as targeted marketing, premium customer acquisition, and financial services, where false positive predictions can lead to unnecessary costs and inefficient resource allocation.

Data Used

The project utilized the **UCI Adult Census Income** dataset obtained from OpenML. The dataset contains **48,842** records describing demographic, educational, occupational, and financial characteristics of individuals. The target variable was encoded as:

- **0:** Income $\leq$ \$50K
- **1:** Income $>$ \$50K

The dataset was split into an **80% training set (39,073 samples)** and a **20% hold-out test set (9,769 samples)** using stratified sampling to preserve the class distribution. The training data contained a moderate class imbalance, with approximately **76.07%** of individuals earning $\leq$ \$50K and **23.93%** earning $>$ \$50K.

To improve predictive performance, several engineered features were incorporated, including age buckets, working-hour categories, capital gain indicators, logarithmic capital gain, higher education status, and an education–hours interaction feature. All preprocessing, feature engineering, imputation, scaling, and categorical encoding were implemented within reusable Scikit-learn pipelines to ensure reproducibility and prevent data leakage.

Final Model and Deployment Artifact

Three ensemble learning algorithms were evaluated using the same preprocessing pipeline:

- Random Forest
- HistGradientBoosting
- Stacking Classifier

Although the Stacking model achieved the highest F1-score and Random Forest achieved the highest ROC AUC, **HistGradientBoosting** was selected as the final production model because it

achieved the **highest Precision**, which directly aligned with the project's business objective of minimizing false positive predictions.

The complete deployment artifact consists of:

- Feature engineering pipeline
- Data preprocessing pipeline
- HistGradientBoosting classifier
- SHAP-based explanation module
- Input validation
- Prediction function
- Joblib serialized model (final_income_prediction_model.joblib)

The deployment pipeline also supports robust inference by validating required inputs and safely handling previously unseen categorical values using `OneHotEncoder(handle_unknown="ignore")`.

Hold-Out Test Performance

The final HistGradientBoosting model achieved the following performance on the untouched hold-out test set:

Metric	Value
Accuracy	0.8164
Precision	0.9875
Recall	0.2357
F1 Score	0.3805
ROC AUC	0.8929
PR AUC	0.7497

The exceptionally high Precision indicates that nearly every individual predicted as earning more than \$50K actually belonged to the positive class. Although Recall is comparatively lower, this trade-off was considered acceptable because minimizing false positives was the primary business requirement.

Selected Decision Threshold

Instead of using the default classification threshold of **0.50**, a threshold of **0.70** was selected following calibration and threshold analysis performed in earlier experiments.

The higher threshold increases the confidence required before predicting the positive class, substantially reducing false positive predictions while maintaining acceptable overall model performance. This threshold directly supports the project's objective of maximizing Precision, even though it results in additional false negatives.

Model Interpretability

Model interpretability was evaluated using both **Permutation Importance** and **SHAP (SHapley Additive Explanations)**.

Permutation Importance identified **capital-gain** as the single most influential feature affecting overall model performance, followed by marital status, capital loss, education level, and the engineered education–hours interaction feature.

SHAP analysis provided both global and local explanations. Globally, the most influential predictors were:

- Marital Status (Married-civ-spouse)
- Capital Gain
- Education Level
- Education–Hours Interaction
- Age
- Capital Loss

Local SHAP explanations demonstrated how these features contributed to individual predictions. High capital gains and being married consistently increased the probability of belonging to the high-income class, while lower education-related contributions reduced prediction confidence for certain individuals. These explanations improve transparency and make the deployed model easier to understand and trust.

Fairness Evaluation

Model fairness was assessed by measuring Precision across demographic groups.

By Sex

Group Precision

Female	0.9896
Male	0.9870

The difference between groups was only **0.26 percentage points**, indicating highly consistent predictive reliability across sexes.

By Race

Race	Precision
Black	1.0000
White	0.9880

Race	Precision
Asian-Pac-Islander	0.9412
Other	1.0000
Amer-Indian-Eskimo	1.0000

No substantial disparities were observed based on Precision alone. However, smaller demographic groups contained relatively few samples, so these estimates should be interpreted cautiously. A comprehensive fairness assessment should additionally monitor Recall, False Positive Rate, False Negative Rate, calibration, and demographic parity during deployment.

Recommended Next Steps

To ensure reliable production performance, the following deployment and monitoring strategy is recommended:

1. **Conduct an A/B Test:** Compare the deployed HistGradientBoosting model against the existing decision process by randomly assigning users to control and treatment groups. Monitor business KPIs such as campaign conversion rate, cost per successful prediction, and false positive rate to validate real-world impact before full deployment.
2. **Continuous Model Monitoring:** Track Precision, Recall, ROC AUC, PR AUC, prediction distribution, inference latency, missing values, unseen categorical values, and data drift in key features such as age, education, workclass, and capital gain. Performance should be reviewed monthly, and retraining should be triggered whenever significant performance degradation or distribution shifts are detected.

By combining high-precision predictions, explainable AI techniques, fairness evaluation, and a reproducible deployment pipeline, the final solution provides a reliable foundation for production use while supporting ongoing monitoring and future model improvements.